

ISYE Course Project – Calimax Grocery Store Chain

For this project, I chose the Calimax success story:

https://www.sas.com/en_us/customers/calimax.html. Calimax is a Mexican grocery chain that dealt with many challenges in managing perishable goods and predicting retail demand. According to the article (and I suppose from common knowledge), “The freshness and quality of perishables can change rapidly – often within days or even hours. Prices fluctuate weekly and margins are tight, adding complexity to accurate forecasting.” This is what Calimax is primarily concerned with, along with reducing their excess inventory.

For this report, I looked to combine predictive modeling, spoilage estimation, and operational optimization using the Weibull distribution, exponential smoothing, and optimization, respectively. Each model works in tandem with the others, and I’ve added in some hypothetical data and examples to try and illustrate how one might go about using the models.

Weibull Distribution: Estimating Spoilage for Perishables

Purpose

When I first read over the story and saw “perishable goods,” my first thought was the Weibull distribution. We learned about the Weibull distribution as a way to estimate the time it takes for something to fail, using the context of machinery. If we think about something “spoiling” as “failing”, then the Weibull distribution is appropriate. It can model spoilage times for perishables, which Calimax might have used to gain insights into when products are likely to become unsellable. This information can be used to optimize inventory rotation (specifically using spoilage time as a constraint) and minimize waste, which we’ll see later on.

Hypothetical Data Requirements

- **Historical spoilage data:** Times to spoilage for key perishables under typical storage conditions. Different foods will have different spoilage times.
- **Environmental conditions:** Temperature, humidity, and other storage factors.
- **Product attributes:** Ripeness, packaging, or other factors affecting spoilage.
- We can get the above information from general common knowledge, but if Calimax wanted to get fancy, they could have used a fractional factorial design to test spoilage using the data above (bonus 4th model for you).

Model Use

- **Inputs:** Empirical spoilage data.
- **Key Parameters**
 - **Shape Parameter (k):** Helps understand how spoilage rate evolves:
 - ($k < 1$): Higher spoilage risk early on (like foods prone to initial contamination).
 - ($k = 1$): Constant spoilage rate (exponential behavior).
 - ($k > 1$): Increasing spoilage risk over time (like fruits ripening).

- **Scale Parameter (λ):** Provides an estimate of the typical spoilage time.
- **Outputs:**
 - Probability of spoilage within a given timeframe.
 - Median and critical spoilage times: (T_{50} , T_{90}) (50% spoil by time T , 90% spoil by time T).
- **Applications:**
 - Calimax could prioritize older stock for immediate sale, assuming it's not past the spoilage time.
 - Calimax could plan markdowns/promotions/sales to reduce waste before spoilage occurs.

Example

For bananas:

- Let's say historical data suggests:
 - (T_{50} , T_{90}) (50% spoil by time T , 90% spoil by time T).
- $\lambda = 8.57$ (scale)
- $k = 2$ (shape)
- Using the Weibull distribution PDF ($f(t) = \frac{k}{\lambda} \left(\frac{t}{\lambda}\right)^{k-1} e^{-\left(\frac{t}{\lambda}\right)^k}$), where $f(t)$ is the likelihood of spoilage occurring at time t , we get:
 - $f(4) \approx 0.233$ (~23% likelihood of spoilage on day 4)
 - $f(7) \approx 0.5$ (~50% likelihood of spoilage on day 7).
- Calimax might promote bananas on day 4 to ensure maximum sales before half of bananas are spoiled. I think in all likelihood they'd want their threshold to be much lower, as seeing half of a food item on shelves spoiled probably isn't the best look for a grocery store.

Exponential Smoothing: Predicting Customer Demand

Purpose

Next, when I read “forecasting demand,” that immediately implied “time series forecasting”. Specifically, we can use Holt-Winters exponential smoothing, as grocery retail data most definitely has seasonality and trends (some foods are in season only part of the year, and some foods might be getting more/less popular in general). We'll focus on perishables again.

Hypothetical Data Requirements

- **Historical sales data:** Daily, weekly, monthly, etc sales for perishables.
- **External/seasonal factors:** Holiday schedules, food seasonality, weather patterns, etc.
- **Store-specific data:** Demand variations across different store locations, which might require individual store forecasts (some foods may be more popular in different regions of Mexico).

Model Integration

- **Inputs:** Historical sales data, including the time-series of past demand and external factors like holidays.
- **Outputs:**
 - **Forecasted demand** for each period (daily, weekly, etc).
 - **Seasonal adjustments:** Capturing recurring patterns in demand, like higher banana sales in summer months.
 - **Trend adjustments:** Understanding whether demand for certain perishables is increasing or decreasing over time.

Applications

There's a ton of stuff Calimax could do with this information. First, they could adjust stock levels to match forecasted demand, making sure the right amount of our perishables is available without overstocking or running out. They could also optimize inventory management to reduce waste from spoilage, especially as demand for perishable goods can fluctuate rapidly. Lastly, they could adjust their marketing strategies, such as promotions or markdowns, based on demand predictions.

Example

For bananas, assume Calimax has weekly sales data and wants to predict the next week's demand. Exponential Smoothing methods, particularly Holt-Winters, can account for both seasonality and trend, and this is specifically we are using "triple" exponential smoothing as opposed to just single/double:

- **Seasonality:** Bananas might see higher sales in summer months or around major holidays, which can be modeled as seasonal variation.
- **Trend:** Sales might have been increasing steadily, showing an upward trend in demand over the past months.

Holt-Winters Model:

1. **Level:** The smoothed value of the time series, adjusted for recent observations.
2. **Trend:** The rate of change in the level over time.
3. **Seasonality:** Adjustments based on the recurring seasonal effects.

Using this model, Calimax could forecast the next period's demand for bananas and account for both the seasonal peaks (e.g., increased demand in the summer) and the general trend (e.g., increasing sales of bananas over time).

Implementation Example:

- If historical data suggests that bananas sell 30% more in the summer months, Holt-Winters would adjust the forecast upward during this period, ensuring more bananas are stocked in advance.
- Conversely, if sales have been declining in the winter months, the model would forecast lower demand, allowing Calimax to reduce stock and avoid overordering.

Optimization: Balancing Inventory, Freshness, and Spoilage, While Minimizing Total Costs

Purpose

Finally, for the third model, I chose optimization. In the article, when it mentioned a key benefit of SAS was “reduced spoilage and inventory expenses,” I thought an optimization model would be most appropriate. Optimization models can help determine the best allocation of inventory across stores while minimizing spoilage, transportation costs, and inventory holding costs, while still meeting demand and reducing overall costs.

Model Setup: Variables, Constraints, and Objective Function

1. **Decision Variables:**
 - x_{ij} : Quantity of product i allocated to store j .
2. **Objective Function:**
 - Minimize total costs:
 - *Total Cost = Spoilage Cost + Transportation Cost + Inventory Holding Cost.*
3. **Constraints:**
 - **Demand Satisfaction:** $\sum_j x_{ij} \geq D_{ij}$, for all i, j
 - Ensure demand D_{ij} for product i at store j is met.
 - **Storage Capacity:** $\sum_i x_{ij} \leq S_j$, for all j
 - Inventory at store j cannot exceed storage capacity S_j .
 - **Transportation Capacity:** $\sum_{i,j} x_{ij} \leq T$
 - Total transported goods must stay within fleet capacity T .
 - **Spoilage Time:** $x_{ij} * t_i \leq ShelfLife_{ij}$, for all i, j
 - t_i : Time product i spends in transit and on shelves.
 - $ShelfLife_{ij}$: Shelf life of product i at store j , derived from Weibull outputs (e.g., T_{50}).

Example

- Exponential smoothing predicts daily demand of 20 units per store for some time period.
- Weibull estimates spoilage risk with $T_{23} = 4$ days.
- Optimization ensures costs are minimized while taking into account:
 - Daily delivery schedules to prevent spoilage.
 - Prioritization of stores with higher demand

Model Refresh Frequency and Combined Workflow

1. Data Refresh:

- **Weibull:** Could run this seasonally as Calimax prepares for stocking seasonal perishables
- **Exponential Smoothing:** Weekly updates using the most recent sales data to capture any shifts in trends or seasonal demand.
- **Optimization:** Daily or weekly updates as demand forecasts and inventory levels change.

2. Hypothetical Workflow for Calimax:

- **Step 1:** Calimax uses Exponential Smoothing (e.g., Holt-Winters) to forecast daily or weekly demand for perishables, incorporating both trend and seasonality.
- **Step 2:** Calimax uses Weibull to estimate spoilage probabilities and shelf life.
- **Step 3:** Calimax runs the optimization model to allocate inventory efficiently across stores, ensuring delivery schedules align with predicted demand and spoilage risk, all while minimizing costs.